

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B	A,B,C	A,B,C	A,B	C,D	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	C	D	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	15.00	20.00	21.00	24.00	70.00	1.01	1.50	40.00

PART-2 : CHEMISTRY

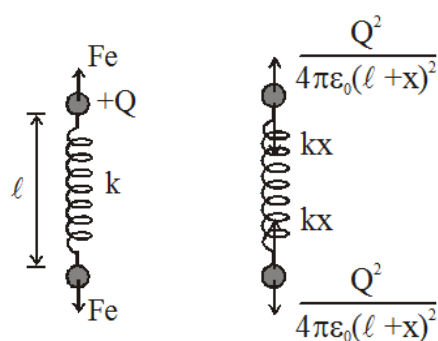
SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	D	B,C,D	A,B,D	A,B	A,B,C	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	C	D	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	6.00	2.00	1.33	382.00 to 384.00	3.00	4.00	7.00	58.00 to 61.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	C	A,D	A,D	A,B,C	C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	D	D	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	2.00	3.66 to 3.67	1.50	8.00	0.40	1.01	0.53	0.75

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

 1. **Ans (A,B)**
 $a_{\text{com}} = g$ (as electrostatic force and spring force is internal forces)

 In frame of COM mg is balanced by Pseudo force.

 At maximum distance between particle $v = 0$

$$\frac{1}{2} kx^2 = \frac{Q^2}{4\pi\epsilon_0} \left[\frac{1}{\ell} - \frac{1}{\ell+x} \right] = \frac{Q^2 x}{4\pi\epsilon_0 \ell(\ell+x)}$$

$$2\pi\epsilon_0(\ell+x) kx = Q^2$$

$$\ell x + x^2 = \frac{Q^2}{2\pi\epsilon_0 k\ell}$$

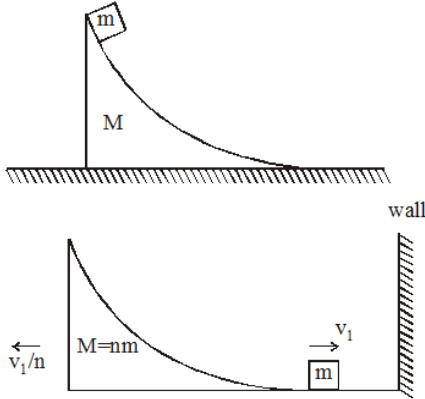
$$\left(x + \frac{\ell}{2}\right) = \frac{Q^2}{2\pi\epsilon_0 k\ell} + \frac{\ell}{4}$$

$$x = \sqrt{\frac{Q^2}{2\pi\epsilon_0 k\ell} + \frac{\ell}{4}} - \frac{\ell}{2} = \sqrt{\frac{3\ell^2}{4} + \frac{\ell^2}{4}} - \frac{\ell}{2}$$

$$\Rightarrow \text{maximum distance} = \frac{3\ell}{2}$$

System is oscillatory but not SHM as net force is not directly proportional to displacement from mean position.

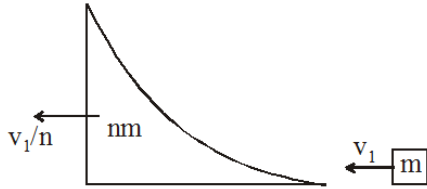
2. Ans (A,B,C)



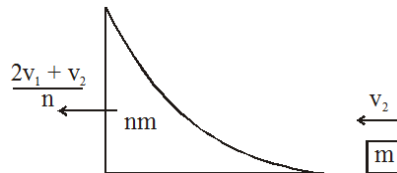
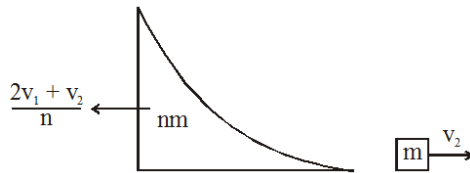
$$p_i = p_f = 0 \Rightarrow V_{com} = 0$$

$$mgh = \frac{1}{2}mv_1^2 + \frac{1}{2}nm \left(\frac{v_1^2}{n^2} \right)$$

$$2gh = v_1^2 \left(1 + \frac{1}{n} \right); v_1 = \left(\frac{2gh}{1 + \frac{1}{n}} \right)$$



$$V_{com} = \frac{nm \left(\frac{v_1}{n} \right) + mv_1}{nm + m} = \frac{2v_1}{n+1} = \frac{2}{n+1} \sqrt{\frac{2gh}{1 + \frac{1}{n}}}$$



$$v_1 - \frac{v_1}{n} = \frac{2v_1 + v_2}{n} + v_2$$

$$v_1 - \frac{3v_1}{n} = v_2 \left(1 + \frac{1}{n} \right)$$

$$v_2 = \frac{n-3}{n+1} v_1$$

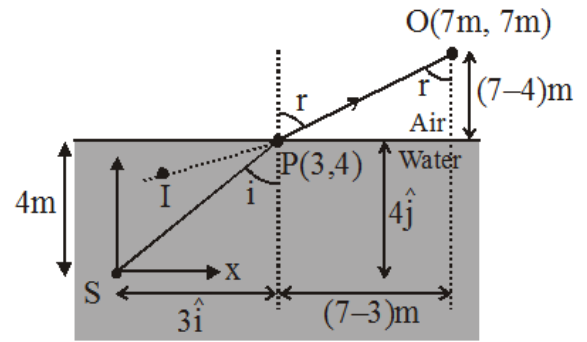
$$V_{com} = \frac{2v_1 + v_2 + v_2}{n+1} = \frac{2}{n+1} \left[v_1 + \frac{n-3}{n+1} v_1 \right]$$

$$= \frac{2}{(n+1)^2} [2n-2] v_1$$

$$= \frac{4(n-1)}{(n+1)^2} v_1 = \frac{4(n-1)}{(n+1)^2} \sqrt{\frac{2gh}{1 + \frac{1}{n}}}$$

$$= \frac{4(n-1)}{(n+1)^2} \sqrt{\frac{2ghn}{n+1}} = \sqrt{\frac{32ghn(n-1)^2}{(n+1)^5}}$$

3. Ans (A,B,C)



$$\tan i = \frac{3m}{4m}; i = 37^\circ$$

$$\tan r = \frac{4m}{3m} = \frac{4}{3}; r = 53^\circ$$

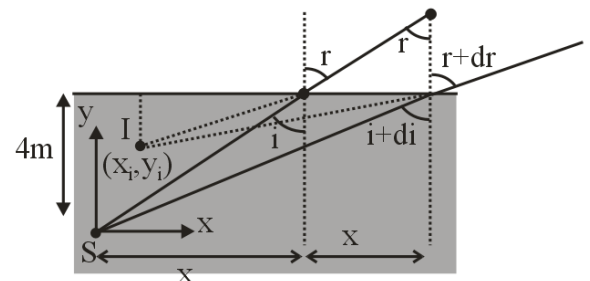
$$\mu_o = \frac{\sin r}{\sin i} = \frac{4}{3}$$

$\Rightarrow$ option A is correct as image will be at line of

refracted ray

$$y - 4 = \tan(90^\circ - r)(x - 3) = \frac{3}{4}(x - 3)$$

$$y = \frac{3}{4}x - \frac{9}{4} + 4 = \frac{3x + 7}{4}$$



$$\tan i = \frac{x}{4m}; \sec^2 i di = \frac{dx}{4m}$$

$$\tan r = \frac{x - x_i}{4 - y_i}; \sec^2 r dr = \frac{dx}{4 - y_i}$$

$$\Rightarrow \frac{\cos^2 r}{\cos^2 i} \frac{di}{dr} = \frac{4 - y_i}{4}$$

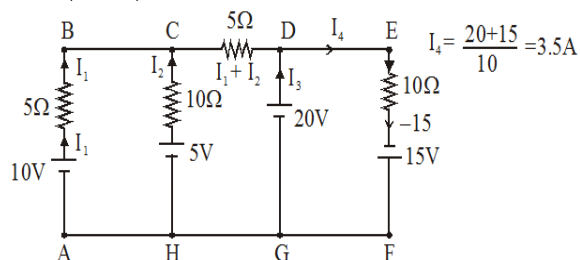
$$\Rightarrow \frac{\cos^3 r}{\cos^3 i} \times \frac{3}{4} = \frac{4 - y_i}{4}$$

$$\left(\frac{3}{5} \right)^3 \times 3 = 4 - y_i; y_i = 4 - \frac{81}{64} = \frac{175}{64}$$

$$\tan r = \frac{x - x_i}{4 - y_i} \Rightarrow \frac{4}{3} = \frac{3 - x_i}{\frac{81}{64}}$$

$$\Rightarrow 3 - x_i = \frac{27}{16}; x_i = \frac{21}{16}$$

4. Ans (A,B)



$$\frac{10 - v}{5} + \frac{5 - v}{10} = \frac{v - 20}{5}$$

$$v = 13 \text{ V}$$

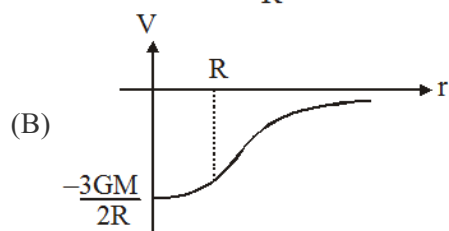
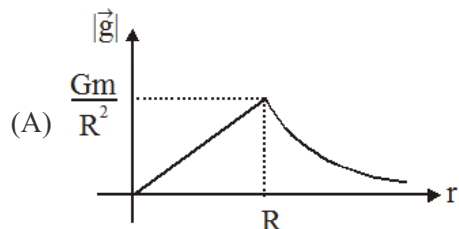
$$I_1 = \frac{-3}{5} \text{ A} = -0.6 \text{ A}$$

$$I_2 = \frac{-8}{10} \text{ A} = -0.8 \text{ A}$$

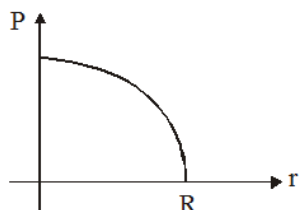
$$I_1 + I_2 = -1.4 \text{ A, - from D to C}$$

$$I_3 = I_4 - I_1 - I_2 = 3.5 + 1.4 \text{ A} = 4.9 \text{ A}$$

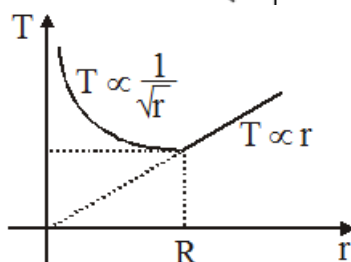
5. Ans (C,D)



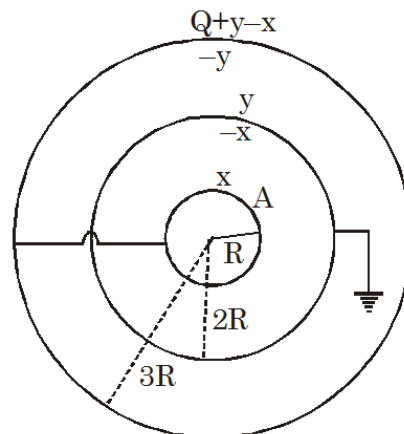
$$(C) P = \frac{3GM^2}{g\pi R^4} \left(1 - \frac{r^2}{R^2}\right)$$



$$(D) T = 2\pi\sqrt{\frac{\ell}{g}}; G \begin{cases} \frac{GM}{R^3}r & r \leq R \\ \frac{GM}{r^2} & r > R \end{cases}$$



6. Ans (A,B,C,D)



$$V_B = \frac{kx}{2R} + \frac{k(y-x)}{2R} + \frac{k(Q-x)}{3R} = 0$$

$$\frac{y}{2R} + \frac{Q-x}{3R} = 0$$

$$3y + 2Q - 2x = 0$$

$$2x - 3y = 2Q \dots\dots(1)$$

$$V_A = V_C$$

$$\frac{kx}{R} + \frac{k(y-x)}{2R} + \frac{k(Q-x)}{3R} = \frac{kx}{3R} + \frac{k(y-x)}{3R} + \frac{k(Q-x)}{3R}$$

$$\frac{x}{R} + \frac{y-x}{2R} = \frac{x}{3R} + \frac{y-x}{3R}$$

$$6x + 3y - 3x = 2x + 2y - 2x$$

$$3x + 3y = 2y$$

$$3x = -y \dots\dots(2)$$

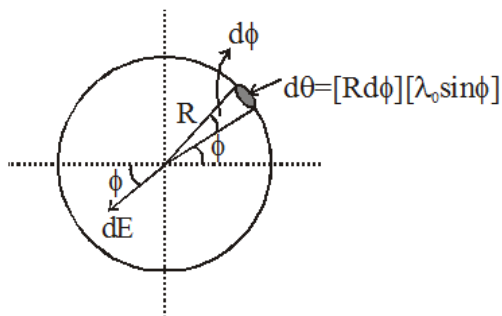
from (1) and (2)

$$2x + 9x = 2Q \Rightarrow x = \frac{2Q}{11} \text{ and } y = -\frac{6Q}{11}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (B)



$$E_x = \int dE_x = - \int dE \cos \phi$$

$$= - \int_0^{2\pi} \frac{K}{R^2} [\lambda_0 R \sin \phi d\phi] \cos \phi$$

$$E_x = - \frac{K\lambda_0}{2R} \left[-\frac{\cos 2\phi}{2} \right]_0^{2\pi} = 0$$

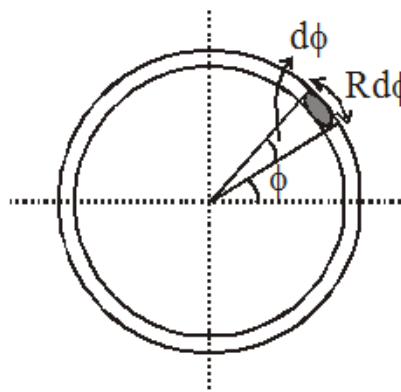
$$E_y = \int dE_y = - \int dE \sin \phi = - \int_0^{2\pi} \frac{K}{R^2} [\lambda_0 R \sin^2 \phi] d\phi$$

$$= - \frac{K\lambda_0}{R} \int_0^{2\pi} \left(\frac{1 - \cos 2\phi}{2} \right) d\phi$$

$$= - \frac{K\lambda_0}{2R} \left[2\pi - \left(\frac{\sin 2\phi}{2} \right)_0^{2\pi} \right]$$

$$E_y = - \frac{\lambda_0}{4\epsilon_0 R}$$

8. Ans (C)



$$d\theta = \lambda(Rd\phi)$$

$$= \lambda_0 R \sin \phi d\phi$$

$$\theta = \int d\theta = \int_0^{2\pi} \lambda_0 R \sin \phi d\phi$$

$$\Rightarrow \vec{F}_{\text{net}} = Q\vec{E} = 0$$

$$\vec{p} = \int \vec{r} d\theta = \int_0^{2\pi} \lambda_0 R^2 [\cos \phi \sin \phi \hat{i} + \sin^2 \phi \hat{j}] d\phi$$

$$= \pi \lambda_0 R^2 \hat{j}$$

From energy conservation

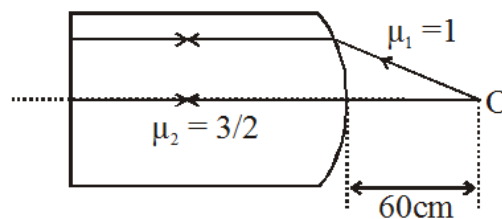
$$KE_i + U_i = KE_f + U_f$$

$$0 - pE \cos 90^\circ = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 - pE \cos \theta$$

$$mv^2 = [\pi \lambda_0 R^2][E_0]$$

$$v = \sqrt{\frac{\pi \lambda_0 E_0}{m}} R$$

9. Ans (D)



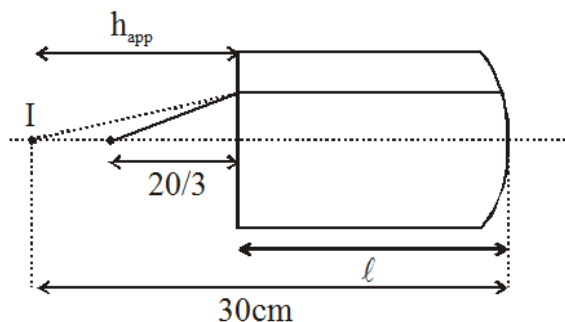
$$\frac{u_2}{V} - \frac{u_1}{u} = \frac{u_2 - u_1}{R}$$

$$\frac{3/2}{\infty} - \frac{1}{-60} = \frac{3/2 - 1}{+R}$$

$$\frac{1}{60} = \frac{1}{2R}$$

$$R = +30\text{cm}$$

10. Ans (C)



$$h_{app} = \frac{20/3}{1} \times \frac{3}{2} = 10\text{cm}$$

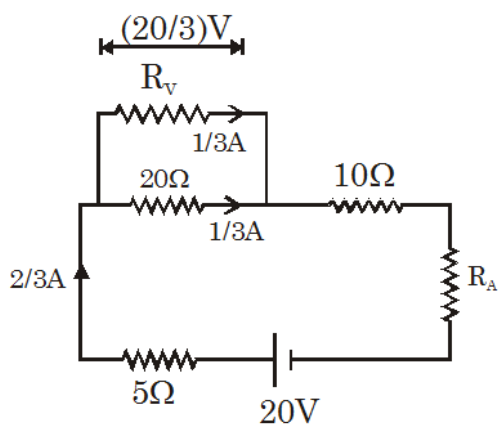
$$h_{app} + \ell = 30\text{cm}$$

$$\ell = 20\text{cm}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (15.00)



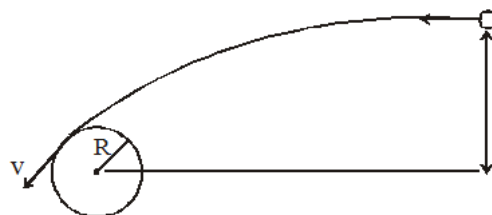
$$R_V = 20\Omega$$

$$\frac{2}{3}\text{A} = \frac{20\text{V}}{5 + 10 + 10 + R_A}$$

$$R_A = 5\Omega$$

$$R_V - R_A = 15\Omega$$

2. Ans (20.00)



$$v_e = \sqrt{\frac{2GM}{R}}$$

$$mv_0 = mvr \dots (i)$$

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv^2 - \frac{GMm}{r}$$

$$v_0^2 = v^2 - \frac{2GM}{r} = \frac{v_0^2 d^2}{r^2} - \frac{2GM}{r}$$

$$d = d_{\max}; r_{\max} R$$

$$v_0^2 = \frac{v_0^2 d^2}{r^2} - \frac{v_e^2 R}{R}$$

$$\frac{v_0^2 d^2}{R^2} = v_0^2 + v_e^2$$

$$d = R\sqrt{1 + \left(\frac{v_e}{v_0}\right)^2}$$

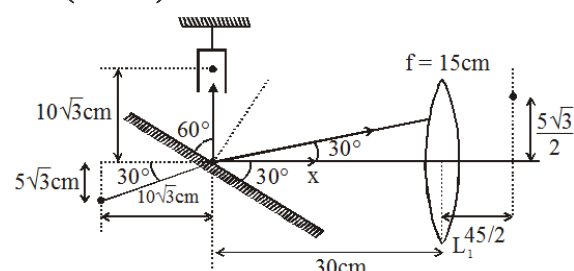
$$m = \rho(\pi d^2 \ell)$$

$$= \pi \rho \ell R^2 \left\{ 1 + \frac{v_e^2}{v_0^2} \right\} = \pi \left(\frac{4}{300} \rho_p \right) (R^3)(1 + 4)$$

$$= \left(\frac{4}{3} \pi \rho_p R^3 \right) \left(\frac{5}{1000} \right) = \frac{5m}{1000} = \frac{m}{200}$$

$$= \frac{m}{10x} \Rightarrow x = 20$$

3. Ans (21.00)



$$u = -45\text{cm}$$

$$\frac{1}{V} = \frac{1}{f} + \frac{1}{u} = \frac{1}{15} - \frac{1}{45}$$

$$h_0 = -5\sqrt{3}\text{cm}$$

$$v = \frac{3-1}{45}$$

$$h_i = \frac{+v}{u} h_0$$

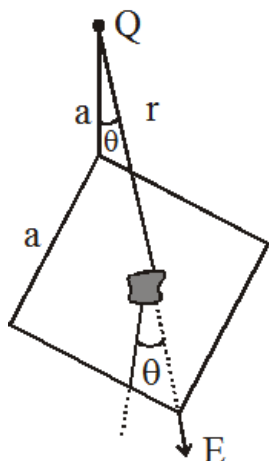
$$v = +\frac{45}{2}\text{cm}$$

$$= \frac{45/2}{-45} \times 5\sqrt{3}$$

$$h_i = +\frac{5\sqrt{3}}{2}$$

$$\frac{\sqrt{3}x}{y} = \frac{105/2}{\left(\frac{5\sqrt{3}}{2}\right)} = 21$$

4. Ans (24.00)



$$d\vec{F} = (\sigma dA)\vec{E}$$

$$dF_{\perp} = (\sigma dA) E \cos \theta$$

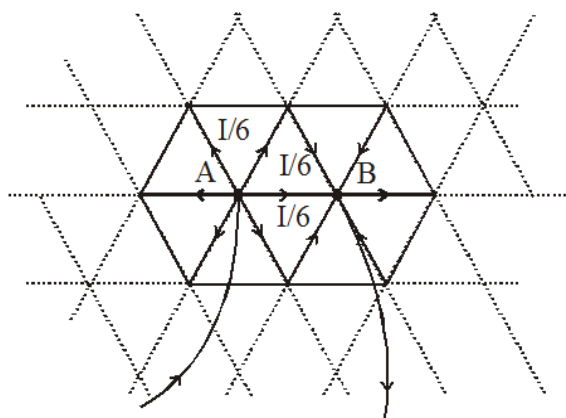
$$\int dF_{\perp} = \sigma \int E dA \cos \theta$$

$$F_{\perp} = \sigma [\text{Electric flux through sheet}]$$

$$F = \sigma \left[\frac{Q}{24\epsilon_0} \right]$$

$$\frac{\sigma Q}{24\epsilon_0} = 24$$

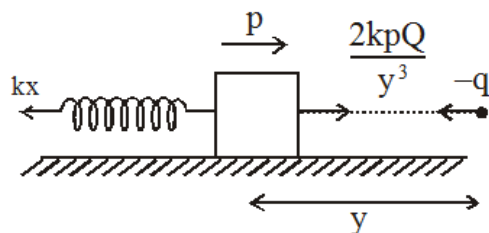
5. Ans (70.00)



$$IR_q = \left(\frac{I}{6} + \frac{I}{6} \right) R$$

$$R_{eq} = \frac{R}{3} = \frac{210}{3} = 70\Omega$$

6. Ans (1.01)



$$kx = \frac{pQ}{2\pi\epsilon_0 y^3} \dots(i)$$

$$F_R = \frac{pQ}{2\pi\epsilon_0 y^3} - kx$$

$$\frac{pQ}{2\pi\epsilon_0 (y - \Delta y)^3} \leq k(x + \Delta y)$$

$$\frac{pQ}{2\pi\epsilon_0 y^3} \left[1 + \frac{3\Delta y}{y} \right] \geq kx \left[1 + \frac{\Delta y}{x} \right]$$

$$x = \frac{y}{3}$$

$$\frac{ky}{3} = \frac{pQ}{2\pi\epsilon_0 y^3}$$

$$y = \left[\frac{3pQ}{2\pi\epsilon_0 R} \right]^{\frac{1}{4}}$$

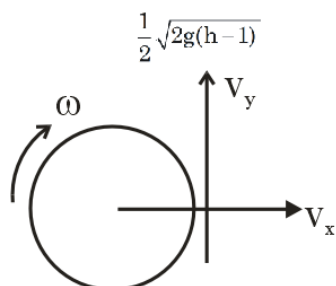
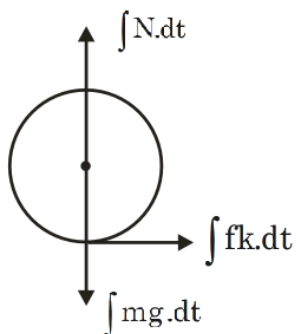
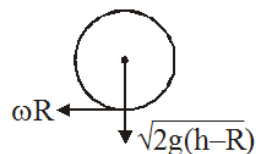
$$\text{Net elongation} = x + y = \frac{y}{3} + y = \frac{4}{3}y$$

$$= \frac{4}{3} \left[\frac{3pQ}{2\pi\epsilon_0 R} \right]^{\frac{1}{4}} = \left[\frac{128pQ}{27\pi\epsilon_0 R} \right]^{\frac{1}{4}}$$

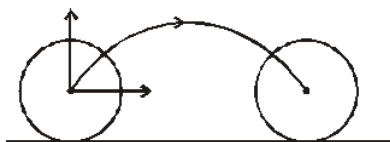
$$a = 128, b = 27$$

$$\frac{a - b}{100} = \frac{101}{100} = 1.01$$

7. Ans (1.50)



$$\int N dt = m \left[\frac{3}{2} \sqrt{2g(h-R)} \right]$$



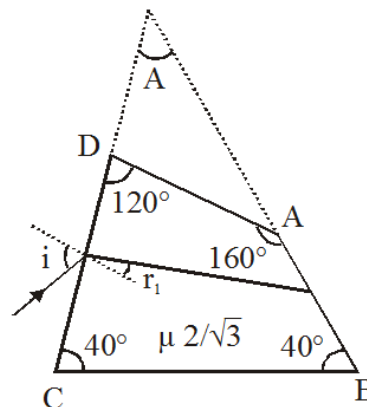
$$\int f k dt = u \int N dt = \frac{3}{2} m \sqrt{2g(h-R)} = m v_x$$

$$v_x = \frac{3}{4} \sqrt{2g(h-R)}$$

$$\text{Range} = \frac{2v_x v_y}{g} = \frac{2 \left(\frac{3}{4} \sqrt{2g(h-R)} \right) \left[\frac{1}{2} \sqrt{2g(h-R)} \right]}{g}$$

$$\frac{3}{2} (2)(h-R) = \frac{3}{2} (h-R)$$

8. Ans (40.00)



$$(1) \sin i = \frac{2}{\sqrt{3}} \sin r_1$$

$$\frac{1}{\sqrt{3}} = \frac{2}{\sqrt{3}} \sin r_1$$

$$r_1 = 30^\circ$$

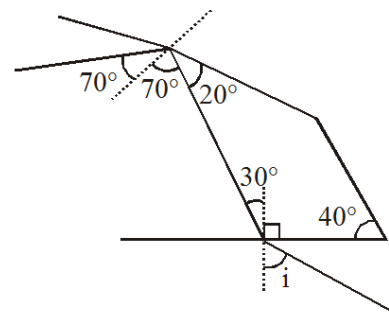
as ray PQ is parallel to side AD prism angle

$$A = 180 - 2(40^\circ) = 100$$

$$r_2 = A - r_1 = 70^\circ$$

$$\frac{2}{\sqrt{3}} \sin r_2 = (1) \sin e$$

$$\sin e = \frac{\sin 70^\circ}{\sin 60^\circ} > 1 \Rightarrow \text{Ray will undergo TIR}$$



$$d_{\text{Net}} = (i - 30^\circ) + \{(180 - 2(70^\circ))\} - (i - 30^\circ) = 40^\circ$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (D)

(A) Ionisation isomer of $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})\text{Cl}_2]\text{Br}$ is $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})\text{ClBr}]\text{Cl}$ and its geometrical isomers are 4

(B) $d_{\text{C-O}}$ (bridging) $>$ $d_{\text{C-O}}$ (terminal)

(C) Di- μ -chloridotetrakis(trichlorostannito)dirhodate(I) ion

(D) $[\text{Co}(\text{NO}_2)_6]^{4-} = d^2 sp^3$

$[\text{CoF}_6]^{3-} = sp^3 d^2$

2. **Ans (B,C,D)**

Option (B) is true for Li_2 and F_2

Option (C) is true for C_2

Option (D) is true for B_2

3. **Ans (A,B,D)**

$$(N \times V)_{\text{I}_2} = (2 \times 1) \times 250 = 500$$

$$\text{n. f. for } \text{Cu}_2\text{S} = 10, N \times V = (0.5 \times 10) \times 100 = 500$$

$$\text{n. f. for } \text{CuS} = 8, N \times V = (0.5 \times 8) \times 125 = 500$$

$$\text{n. f. for } \text{S}_2\text{O}_3^{2-} = 1, N \times V = (0.5 \times 1) \times 1000 = 500$$

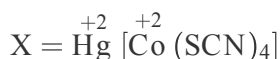
$$\text{n. f. for } \text{I}_2 = 2$$

PART-2 : CHEMISTRY

SECTION-I (ii)

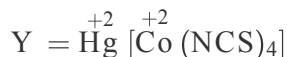
7. **Ans (C)**

$\text{HgI}_2 \rightarrow$ scarlet red precipitate



$$\text{hyb} = \text{sp}^3$$

$$n = 3$$



$$\text{hyb} = \text{sp}^3$$

$$n = 3$$

8. **Ans (C)**

$\text{Hg}[\text{Co}(\text{SCN})_4]$		$\text{Hg}[\text{Co}(\text{NCS})_4]$	
$(\text{SCN})^-$	$(\text{NCS})^-$	$(\text{NCS})^-$	$(\text{SCN})^-$
4	0	4	0
3	1	3	1
2	2	2	2
1	3	1	3
0	4	0	4
total = 5		total = 5	

$$\text{sum} = 5+5 = 10$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (6.00)**

Option (R) and (W) are incorrectly matched.

2. **Ans (2.00)**

(1) It is an endothermic reaction

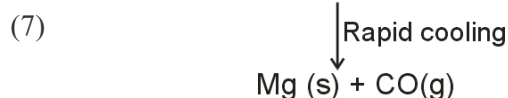
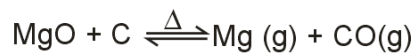
(2) If menite = $\text{FeO} \cdot \text{TiO}_2$

(3) KCN act as depressant

(4) $\text{Zn} \rightarrow$ distillation

(5) Complex formed is $\text{Na}[\text{Au}(\text{CN})_2]$

(6) Leaching of red bauxite



3. **Ans (1.33)**

$$n - 1 = 3 \Rightarrow n = 4$$

$$\text{slope} = (n - 1) \times 4K = \tan 45$$

$$12K = 1 \Rightarrow K = \frac{1}{12}$$

$$r = K[\text{A}]^4$$

$$= \frac{1}{12} \times (2)^4$$

$$= 1.33 \text{ M/sec.}$$

4. **Ans (382.00 to 384.00)**

After long time, total pressure is 617 mm Hg.

$$5P = 617 - 32.5$$

$$2P = 233.8 \dots (1)$$

$$\text{After 30 minute, } 2P + 3x + 4x = 317 - 32.5$$

$$2x = 33.8 \dots (2)$$

The expression for the rate of the reaction is

$$k = \frac{2.303}{t} \log \frac{a}{a-x} = \frac{2.303}{30} \log \frac{233.8}{233.8 - 33.8} \dots (3)$$

After 75 minute,

$$k = \frac{2.303}{t} \log \frac{a}{a-x}$$

$$= \frac{2.303}{(75)} \log \frac{233.8}{233.8 - 2y} \dots (4)$$

$$\text{Total pressure } 2P + 3y + 32.5$$

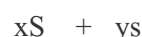
$$= 383.2 \text{ mm Hg}$$

5. **Ans (3.00)**

$$m = \frac{3.12 \times 10^{-3}}{0.52} = 6 \times 10^{-3} \text{ mole / kg}$$

$$\cong \text{molarity} \cong 6 \times 10^{-3} \text{ mole / L}$$

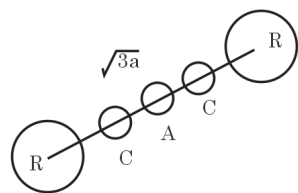
$$S = 2 \times 10^{-3} \text{ mole / L}$$



$$\text{total molarity} = (x + y)s = 6 \times 10^{-3}$$

$$x + y = 3$$

8. Ans (58.00 to 61.00)



$$\begin{aligned} & \frac{2R + 4r_{C+}}{\sqrt{3}a} \times 100 \\ &= \frac{2R + 4 \times 0.225R}{\sqrt{3}} \times 100 \\ &= \frac{2.9}{\sqrt{3}} \times \frac{\sqrt{2}a}{4} \times 100 \\ &= \frac{2.9}{29} \times \frac{24}{4} \times 100 = 60 \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C)

$$F(x) = \int P(x) dx$$

$$\Rightarrow F(x) = x^9 - 3x^6 - 5x^4 + 15x + C$$

$$F(x) = x(x^5 - 5)(x^3 - 3) + C$$

Now use Rolle's theorem on $F(x)$ in $[0, 5^{1/5}]$, $[5^{1/5}, 3^{1/3}]$ and $[0, 3^{1/3}]$

2. Ans (C)

$$f'(-1) = 0, f(-1) = 10$$

$$f(0) = 5, f''(1) = 0$$

$$f''(x) = a(x-1)$$

$$f'(x) = a \left(\frac{x^2}{2} - x \right) + c$$

$$x = -1, f'(-1) = \frac{3a}{2} + c = 0 \quad \dots(1)$$

$$f(x) = a \left(\frac{x^3}{3} - \frac{x^2}{2} \right) + cx + d$$

$$f(0) = 5 \Rightarrow d = 5$$

$$f(-1) = 10 \Rightarrow \frac{2a}{3} + c = -5 \quad \dots(2)$$

from (1) and (2) $a = 6, c = -9$

$$f(x) = x^3 - 3x^2 - 9x + 5$$

$$f(0) = 5 \quad 0 < \alpha < 1$$

$$f(1) = -6 \quad [\alpha] = 0$$

$$f(-2)f(-3) < 0$$

$$-3 < \beta < -2$$

$$[\beta] = -3$$

$$f(4)f(5) < 0$$

$$4 < \gamma < 5$$

$$[\gamma] = 4$$

3. Ans (A,D)

$$\lim_{\alpha \rightarrow 0} \frac{\alpha^2}{(\sec \alpha)^{1/x} - (\sec \alpha)^{1/y}} = \frac{0}{0} \text{ form}$$

apply L-Hopital Rule

$$\lim_{\alpha \rightarrow 0} \frac{2\alpha}{\frac{1}{x}(\sec \alpha)^{1/x} \tan \alpha - \frac{1}{y}(\sec \alpha)^{1/y} \tan \alpha}$$

$$f(x, y) = \frac{2}{\frac{1}{x} - \frac{1}{y}} = \frac{2xy}{y - x}$$

$$\Rightarrow f(2, 3) = 12, \frac{2xy}{y - x} = 12 \Rightarrow xy = 6y - 6x$$

$$y = \frac{6x}{6 - x}$$

$$\text{differentiate both sides } \frac{dy}{dx} = 6 \left\{ \frac{(6-x) + x}{(6-x)^2} \right\}$$

$$\frac{dy}{dx} = \frac{36}{(6-x)^2} \Rightarrow \frac{dy}{dx} \Big|_{x=2} = \frac{36}{16} = \frac{9}{4}$$

$$\frac{36}{(6-x)^2} \text{ can't be zero } \Rightarrow \text{No horizontal tangent}$$

4. Ans (A,D)

$$y^x - x^y = 1$$

$$\text{at } x = 1 \Rightarrow y = 2$$

differentiate

$$x \cdot y^{x-1} \cdot \frac{dy}{dx} + y^x \cdot \ln y - y \cdot x^{y-1} - x^y \cdot \ln x \cdot \frac{dy}{dx} = 0$$

$$\text{put } x = 1 \text{ \& } y = 2$$

$$\frac{dy}{dx} + 2 \ln 2 - 2 = 0$$

$$\Rightarrow \frac{dy}{dx} = 2(1 - \ln 2)$$

$$\Rightarrow \ln \left(\frac{e^2}{4} \right)$$

6. Ans (C)

$$\frac{f'(x)}{f(x)} = \frac{{}^nC_0}{{}^nC_0x+1} + \frac{{}^nC_1}{{}^nC_1x+2} + \dots + \frac{{}^nC_n}{{}^nC_nx+n+1}$$

$$\Rightarrow \frac{f'(0)}{f(0)} = \frac{{}^nC_0}{1} + \frac{{}^nC_1}{2} + \dots + \frac{{}^nC_n}{n+1}$$

$$f'(0) = (n+1)! \left(\frac{2^{n+1} - 1}{n+1} \right) = n!(2^{n+1} - 1)$$

$$\therefore n!(2^{n+1} - 1) = \frac{3}{2} 7!$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (B)**

$$\begin{aligned} \text{Let } f(x) &= (x-1)^2(ax^2 + bx + c) \\ \Rightarrow a + b + c &= 2 \text{ \& } f'(c) = -8, f'(2) = 8 \\ \Rightarrow c &= 3, a = 1, b = -2 \\ \Rightarrow f(x) &= (x-1)^2(x^2 - 2x + 3) \\ g(x) &= 3(x-1)^2 \Rightarrow g'(0) = -6, f'(0) = -8 \\ \text{Angle of intersection} \end{aligned}$$

$$= \tan^{-1} \left(\frac{2}{1+48} \right) = \tan^{-1} \frac{2}{49}$$

8. **Ans (D)**

$$\begin{aligned} \text{Length of subtangent} &= \left| \frac{y}{dy/dx} \right| = \left| \frac{y}{6(x-1)} \right| \\ \left| \frac{3(x-1)^2}{6(x-1)} \right| &= 2018 \Rightarrow \frac{x-1}{2} = \pm 2018 \\ x &= 4037 \text{ or } x = -4035 \end{aligned}$$

9. **Ans (D)**

$$\because \lim_{x \rightarrow \infty} \frac{f(x)}{|x|^3} = 0 \Rightarrow f(x) \text{ is either linear or quadratic}$$

$$\begin{aligned} \text{Let } f(x) &= ax^2 + bx (\because f(0) = 0) \\ \therefore \lim_{x \rightarrow \infty} \left(\sqrt{f(x)} - x \right) &= \lim_{x \rightarrow \infty} \frac{(a-1)x^2 + bx}{x \left(\sqrt{a + \frac{b}{x} + 1} \right)} = -1 \\ \Rightarrow a - 1 &= 0 \text{ and } \frac{b}{\sqrt{a} + 1} = -1 \Rightarrow b = -2 \end{aligned}$$

$$\text{Hence, } f(x) = x^2 - 2x$$

Now proceed.

10. **Ans (B)**

$$\because \lim_{x \rightarrow \infty} \frac{f(x)}{|x|^3} = 0 \Rightarrow f(x) \text{ is either linear or quadratic}$$

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$$\text{Hence, } f(x) = x^2 - 2x$$

Now proceed.

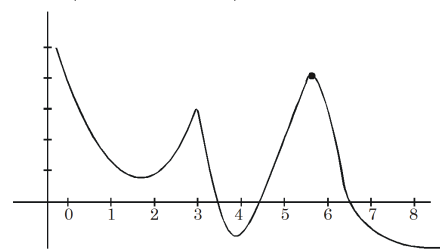
PART-3 : MATHEMATICS

SECTION-II

1. **Ans (2.00)**

$$\begin{aligned} \text{If } (h, k) &\text{ is point of contact of tangent,} \\ \Rightarrow \frac{dy}{dx} &= \frac{-\sin(h+2k)}{1+2\sin(h+2k)} = \frac{-1}{3} \\ \Rightarrow \sin(h+2k) &= 1 \text{ and } k = \cos(h+2k) \\ \Rightarrow k &= 0 \text{ and } \sinh = 1 \end{aligned}$$

2. **Ans (3.66 to 3.67)**



$$\begin{aligned} f'(x) &= f'(x) (f(x))^2 \\ \Rightarrow f'(x) &= 0, f(x) = 1, f(x) = -1 \\ f'(x) &= 0 \text{ for atleast 4 points} \\ f(x) &= 1 \text{ for atleast 5 points} \\ f(x) &= -1 \text{ for atleast 2 point} \\ \text{so } f'(x) &= f'(x) (f(x))^2 \text{ for atleast 11 points.} \end{aligned}$$

3. **Ans (1.50)**

$$\begin{aligned} \text{Put } y &= 1 : 2 f(x) = f(x) + 3^x \Rightarrow f(x) = 3^x \\ \lim_{x \rightarrow 0} \frac{3^x + 3^{-x} - 2}{x^2} &= (\ln 3)^2 \end{aligned}$$

4. **Ans (8.00)**

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{x^3 (\sqrt{1+x^4} - x^2)}{(\sqrt{x^2 + \sqrt{1+x^4}} + \sqrt{2x})} \\ \lim_{x \rightarrow \infty} \frac{x^3}{(\sqrt{x^2 + \sqrt{1+x^4}} + \sqrt{2x}) (\sqrt{1+x^4} + x^2)} \\ \lim_{x \rightarrow \infty} \frac{1}{\left(\sqrt{1 + \sqrt{\frac{1}{x^4} + 1}} + \sqrt{2} \right) \left(\sqrt{\frac{1}{x^4} + 1} + 1 \right)} \\ = \frac{1}{4\sqrt{2}} = \frac{\sqrt{2}}{8} \end{aligned}$$

5. **Ans (0.40)**

$$\begin{aligned} \text{Let } x &= \frac{1}{t} \\ \lim_{t \rightarrow 0} \left(\frac{f(3+3t)}{f(3)} \right)^{\frac{1}{t}} &= e^{\lim_{t \rightarrow 0} \left(\frac{f(3+3t)-f(3)}{t f(3)} \right)} \\ &= e^{\lim_{t \rightarrow 0} \left(\frac{f'(3+3t) \cdot 3}{f(3)} \right)} = e \end{aligned}$$

6. **Ans (1.01)**

$$f(x) = \prod_{r=1}^{100} (x-r)^{r(101-r)}$$

$$\ln(f(x)) = \sum_{r=1}^{100} r(101-r) \ln(x-r)$$

$$\Rightarrow \frac{f'(x)}{f(x)} = \sum_{r=1}^{100} \frac{r(101-r)}{(x-r)}$$

$$\therefore \frac{f'(101)}{f(101)} = \sum_{r=1}^{100} \frac{r(101-r)}{(101-r)} = \sum_{r=1}^{100} r$$

$$= \frac{100 \times 101}{2}$$

$$\therefore P = 50 \times 101$$

$$\therefore \frac{P}{5000} = \frac{50 \times 101}{5000} = 1.01$$

7. **Ans (0.53)**

Let width and height of left compartment is x and y then

$$\text{Volume } V = 3xy$$

$$\text{or } y = \frac{V}{3x}$$

$$\therefore \frac{dy}{dx} = \frac{1}{3} \left[\frac{x \frac{dv}{dt} - v \cdot \frac{dx}{dt}}{x^2} \right]$$

$$= \frac{1}{3} \left[\frac{10.5 - 70.3}{10^2} \right] = \frac{-8}{15} \text{ ft/min}$$

$$\left(\frac{dx}{dt} = 3 \text{ ft/min}, \frac{dv}{dt} = 5 \text{ ft}^3/\text{min} \right)$$

8. **Ans (0.75)**

Put $x = 2, \frac{1}{2}, -1$ successively

$$f(2) + f\left(\frac{1}{2}\right) = 3$$

$$f\left(\frac{1}{2}\right) + f(-1) = \frac{3}{2}$$

$$f(-1) + f(2) = 0$$

$$\text{solving above, } f(2) = \frac{3}{4}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,D	A,C,D	A,B,C	A,C,D	A,C	A,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	B	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.60	3.73 to 3.75	13.50	29.53 to 29.55	0.60	0.67	58.00	2.25

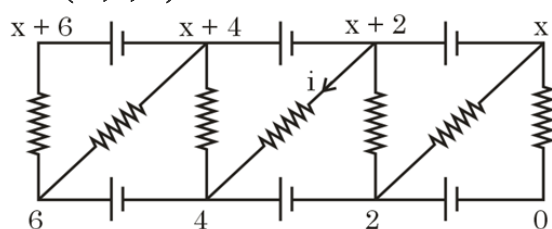
PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C,D	B,C,D	B,D	A,B,C	A,B,D	B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	A	A	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	6.00	3.00	1.00	3.00	4.00	4.00	3.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,D	D	A,B,C,D	A,B,C	B,C	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	C	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.00	3.20	2.00	0.50	9.00	0.20	8.00	3.75

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

 2. **Ans (A,C,D)**


$$\frac{4x}{1} + 3 \left(\frac{x-2}{1} \right) = 0$$

$$4x + 3x - 6 = 0$$

$$x = \frac{6}{7}$$

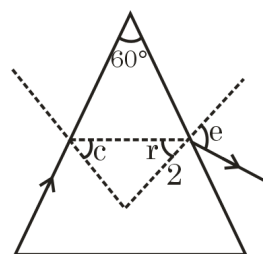
$$i = \frac{x+2-4}{1} = x-2 = \frac{\frac{6}{7}-2}{1} = -\frac{8}{7}$$

 3. **Ans (A,B,C)**

$$\delta_{\min} = 60^\circ = 2 \times 60^\circ - A \Rightarrow A = 60^\circ$$

$$r = \frac{A}{2} = 30^\circ$$

$$\sin 60^\circ = \mu \sin 30^\circ \Rightarrow \mu = \sqrt{3}$$

 For $\delta_{\max}$:


$$r_2 = 60 - C$$

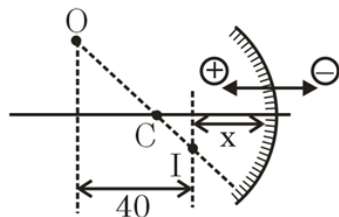
$$\sqrt{3} \sin(60 - C) = \sin e$$

$$\delta_{\max} = (90 + e) - 60^\circ$$

$$= 30^\circ + e$$

$$= 30^\circ + \sin^{-1} \left(\frac{\sqrt{3}}{2} - \frac{1}{2} \right)$$

5. Ans (A,C)



$$u = x + 40$$

$$\mathbf{v} = \mathbf{x}$$

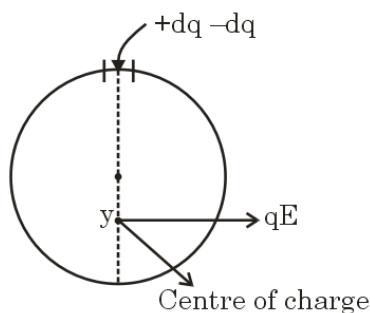
$$m = -\frac{1}{2} = \frac{-x}{x+40}$$

$$2x = x + 40$$

$$\frac{1}{f} = \frac{1}{40} + \frac{1}{80}; \frac{1}{f} = \frac{3}{80}$$

$$f = \frac{80}{3}; x = 40$$

6. Ans (A,D)



Center of charge (COC)

$$y = \frac{Q(0) + (-dq)R}{Q - dq}$$

$$= \frac{-\lambda dR}{\lambda 2\pi R} (Q \gg dq) = -\frac{d}{2\pi}$$

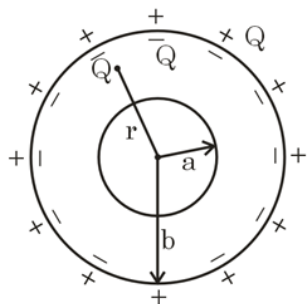
τ about COM

$$\tau = qE \times y = qE \frac{d}{2\pi}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A)

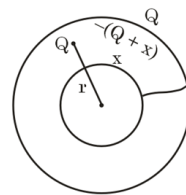


$$Q_1 = 0; Q_2 = Q \Rightarrow \frac{Q_1}{Q_2} = 0$$

$$v_1 = \frac{kQ}{r}; \quad v_2 = \frac{kQ}{b}$$

$$\frac{v_1}{v_2} = \frac{b}{r} = \frac{4}{2} = 2 \text{ and } v_0 = v_1$$

After connecting :



$$V_1' = V_2'$$

$$\frac{kQ}{r} + \frac{kx}{a} - \frac{kx}{b} = \frac{kQ}{b}$$

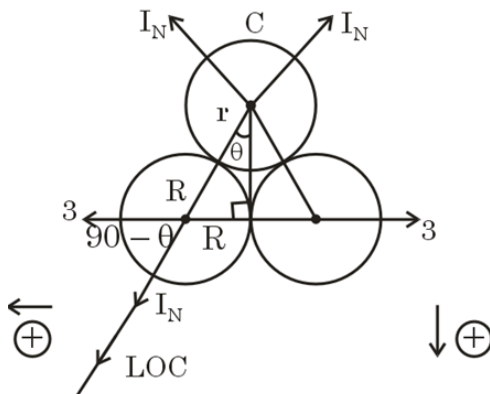
$$k_x \left(\frac{1}{a} - \frac{1}{b} \right) = kQ \left(\frac{1}{b} - \frac{1}{r} \right)$$

$$x = \frac{Q\left(\frac{1}{b} - \frac{1}{r}\right)}{\left(\frac{1}{a} - \frac{1}{b}\right)} = \frac{1 \times \left(\frac{1}{4} - \frac{1}{2}\right)}{\frac{1}{1} - \frac{1}{4}} = -\frac{\frac{1}{4}}{\frac{3}{4}} = -\frac{1}{3}$$

$$Q'_1 = x \text{ and } Q'_2 = Q$$

$$\frac{Q_1'}{Q_2'} = \frac{x}{Q} = \frac{1}{3}$$

8. Ans (B)



$$\sin \theta = \frac{R}{r+R} = \frac{R}{\sqrt{2}R - R + R} = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ$$

For C : $-2I_N \cos\theta = 1(v - 10) \dots\dots(1)$ (IMT in y direction)

For A : $I_N \sin \theta = 2(3 - 0) \dots\dots\dots(2)$ (IMT in x direction)

(1)/(2)

$$-2 = \frac{v - 10}{2 \times 3}$$

$$-12 = v - 10$$

$$v = -2$$

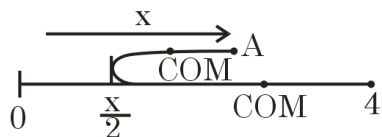
from (2)

$$I_N = \frac{6}{\sin \theta} = 6\sqrt{2}$$

$$I_T = I_N \cos \theta$$

$$= 6\sqrt{2} \times \frac{1}{\sqrt{2}} = 6$$

9. Ans (B)



$$(B) x_{COM} \text{ of moving part} = \frac{x}{2} + \frac{x}{4} = \frac{3x}{4}$$

$$v_{COM} = \frac{3}{4} \frac{dx}{dt} = \frac{3}{4} \times 2 = \frac{3}{2}$$

(A) x_{COM} of part still at rest

$$= \frac{x}{2} + \frac{(4 - \frac{x}{2})}{2}$$

$$= \frac{x}{2} + \left(2 - \frac{x}{4}\right) = 2 + \frac{x}{4}$$

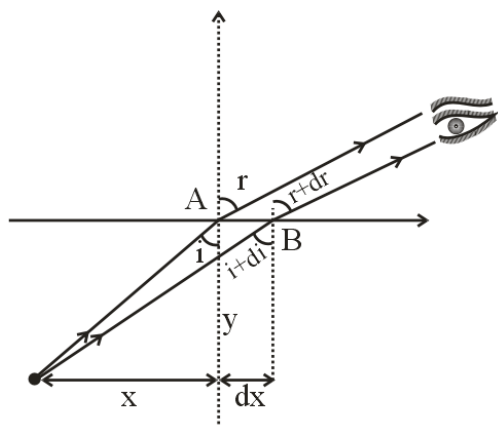
$$v_{COM} = 0 + \frac{1}{4} \frac{dx}{dt} = \frac{2}{4} = \frac{1}{2}$$

(C) Let in time dt dx distance is moved by A

$$\text{then mass added } dm = \left(\frac{dx}{2}\right) \lambda$$

$$\frac{dm}{dt} = \frac{\lambda}{2} \frac{dx}{dt} = \frac{\lambda}{2} v = \frac{0.5}{2} \times 2 = 0.5 \text{ m/s}$$

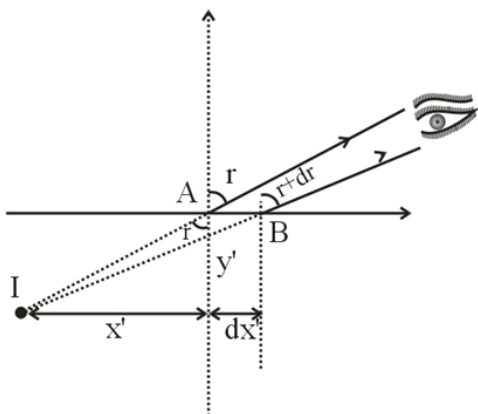
10. Ans (D)



$$\tan i = \frac{x}{y}$$

$$x = y \tan i$$

$$dx = y \sec^2 i di$$



$$\tan r = \frac{x'}{y'}$$

$$x' = y' \tan r$$

$$dx' = y' \sec^2 r dr$$

$$dx = dx' \Rightarrow y \sec^2 i di = y' \sec^2 r dr$$

$$y' = y \frac{\sec^2 i}{\sec^2 r} \frac{di}{dr} = y \frac{\cos^2 r}{\cos^2 i} \frac{di}{dr} \dots (i)$$

From Snell's Law:

$$\mu \sin i = \sin r$$

$$\mu \cos i di = \cos r dr$$

$$\frac{di}{dr} = \frac{\cos r}{\cos i} \frac{1}{\mu}$$

so from eq.(i)

$$y' = \frac{y}{\mu} \left(\frac{\cos^2 r}{\cos^2 i} \right)$$

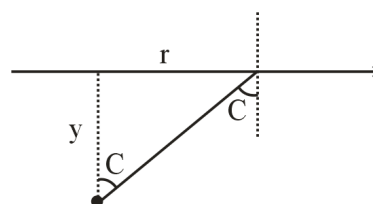
For (P)

$$y' = \frac{8}{\frac{4}{3}} \left(\frac{\cos^2 53^\circ}{\cos^2 37^\circ} \right) = 2.53$$

For (Q)

$$x' - x = y' \tan r - y \tan i$$

For (R)



$$\sin C = \frac{1}{\mu} = \frac{3}{4}$$

$$\tan C = \frac{r}{y} \Rightarrow r = y \tan C = 8 \times \frac{3}{\sqrt{7}}$$

For (S)

$$P_{out} = \frac{P}{4\pi} 2\pi(1 - \cos C) = \frac{P}{2} \left(1 - \frac{\sqrt{7}}{4}\right)$$

$$\frac{P_{out}}{P} \times 100 = 50 \left(1 - \frac{\sqrt{7}}{4}\right)$$

PART-1 : PHYSICS

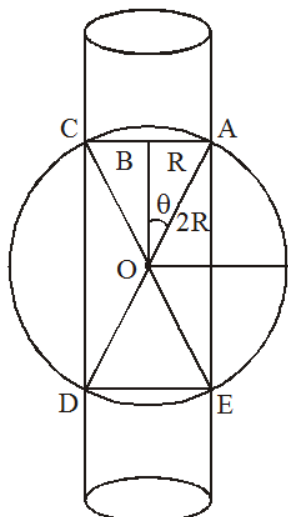
SECTION-II

1. Ans (5.60)

$$\mu \sin \theta = 1 \Rightarrow \sin \theta = \frac{1}{\mu} \Rightarrow \tan \theta = \frac{1}{\sqrt{\mu^2 - 1}}$$

$$\Rightarrow \tan \theta = e^{-x/2d} = \frac{dy}{dx} \Rightarrow y = 2d (1 - e^{-x/2d})$$

2. Ans (3.73 to 3.75)



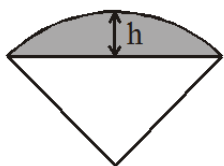
$$\theta = 30^\circ$$

$$OB = 2R \cos \theta = R\sqrt{3}$$

enclosed volume = volume of cylinder ACDE

volume of two spherical cap

vol of a spherical caps



$$V_{\text{cap}} = \pi h^2 \left(R - \frac{h}{3} \right)$$

$$\text{here } h = 2R - 2R \cos \theta$$

$$= 2R - R\sqrt{3}$$

$$\text{So } V_{\text{cap}} = \pi (2R - R\sqrt{3})^2 \left(2R - \frac{2R - R\sqrt{3}}{3} \right)$$

$$= \left(\frac{16}{3} - 3\sqrt{3} \right) \pi R^3$$

$$\text{Total volume} = (\pi R^2) 2R\sqrt{3} + 2 \times \left(\frac{16}{3} - 3\sqrt{3} \right) \pi R^3$$

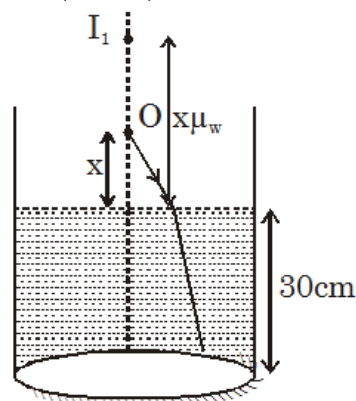
$$= 2\pi R^3 \left(\sqrt{3} + \frac{16}{3} - 3\sqrt{3} \right)$$

$$= 2\pi R^3 \left(\frac{16 - 6\sqrt{3}}{3} \right)$$

$$d = \frac{\rho V}{\epsilon_0} = V = \frac{2}{3} \pi R^3 [16 - 6\sqrt{3}]$$

$$= \frac{2}{3} \pi R \frac{1}{\pi} [16 - 6\sqrt{3}]$$

3. Ans (13.50)



$$\frac{1}{f} = (1.5 - 1) \left(\frac{2}{R} \right)$$

$$f = R = 60$$

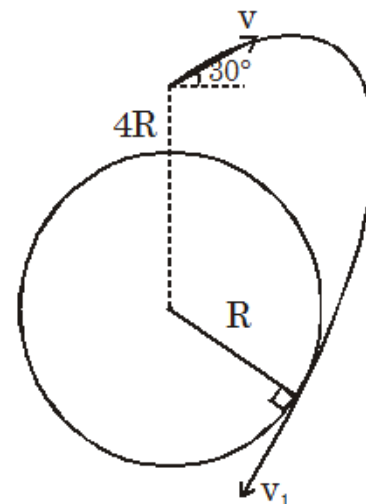
For curved surface refraction

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1.5}{-60} - \frac{4/3}{-(30 + \frac{x^4}{3})} = \frac{1.5 - 4/3}{60}$$

$$x = 13.5$$

4. Ans (29.53 to 29.55)



$$mv \frac{\sqrt{3}}{2} \times 4R = mv_1 R$$

$$v_1 = 2\sqrt{3}v$$

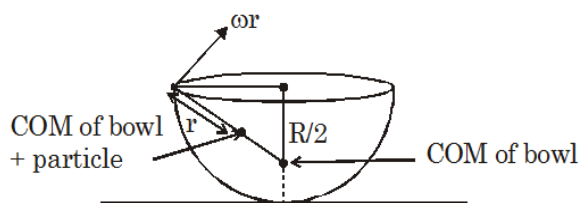
Mechanical energy conservation

$$\frac{1}{2} mv^2 - \frac{GMm}{4R} = \frac{1}{2} m(12v^2) - \frac{GMm}{R}$$

$$\frac{3}{4} \frac{GMm}{R} = \frac{11}{2} mv^2$$

$$v = \sqrt{\frac{3}{22} \frac{GM}{R}}$$

5. Ans (0.60)



Conservation of angular momentum about COM of bowl + particle

$$mu \frac{R}{4} = \left(\frac{25}{24} m R^2 \right) \omega$$

$$w = \frac{6u}{25R}$$

$$v_y = \omega r \cos \theta = \frac{\omega R}{2}$$

$$= \frac{3u}{25} = 0.6 \text{ m/s}$$

6. Ans (0.67)

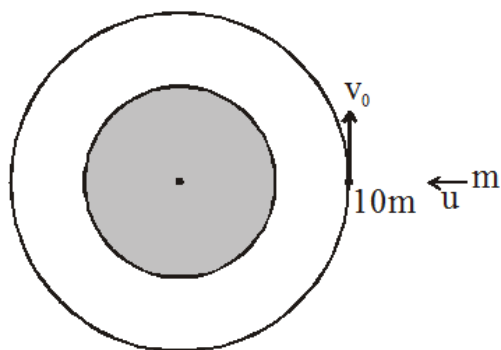
Let $x \rightarrow$ be the number of electrons striking the surface per unit time.

$$F = PA = nmv = I/qmv$$

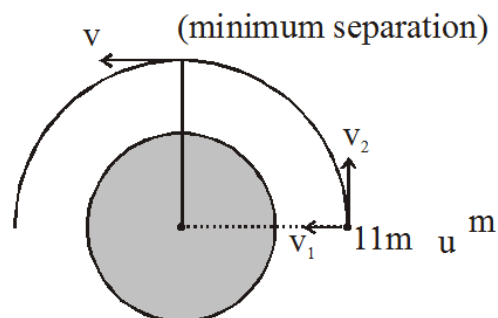
$$I = \frac{PA\rho}{mv}$$

$$I = \frac{9.1 \times 1 \times 10^{-4} \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31} \times 24 \times 10^7} = 0.67 \text{ A}$$

7. Ans (58.00)



Just before



Just after

7. Ans (58.00)

Momentum conservation

$$mu = 11mv_1 \Rightarrow v_1 = \frac{u}{11}$$

$$10m v_0 = 11mv_2 \Rightarrow v_2 = \frac{10v_0}{11}$$

Conservation of angular momentum about centre of planate

$$11m \left(\frac{10}{11} v_0 \right) r = 11mv \frac{r}{2}$$

$$v = \frac{20v_0}{11} \dots (i)$$

From energy conservation

$$\frac{1}{2} (11m) (v_1^2 + v_2^2) - \frac{GM(11m)}{r}$$

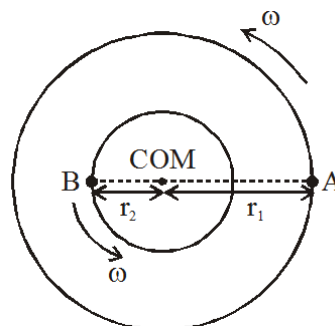
$$= \frac{1}{2} (11m) v^2 - \frac{GM(11m)}{r/2}$$

$$\frac{u^2}{121} + \frac{100v_0^2}{121} - \frac{2GM}{r} = v^2 - \frac{4GM}{r}$$

$$\frac{u^2}{121} = -\frac{2GM}{r} - \frac{100v_0^2}{121} + \frac{400}{121} v_0^2$$

$$u = \sqrt{\frac{58GM}{r}}$$

8. Ans (2.25)



$$\text{Rate of area swept } \frac{dA}{dt} = \frac{L}{2m} = \frac{mr^2\omega}{2m} = \frac{\omega r^2}{2}$$

(r = distance from COM)

$$\Rightarrow \frac{dA}{dt} \propto r^2$$

$$\Rightarrow \frac{\left(\frac{dA}{dt} \right)_A}{\left(\frac{dA}{dt} \right)_B} = \frac{r_1^2}{r_2^2} = \left(\frac{m_2}{m_1} \right)^2 = 2.25$$

PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (A)

$$(P) K = \frac{K_A}{2} = \frac{0.693}{80} s^{-1}$$

$$r = [A]_0 = \frac{0.693}{80} \times 0.8$$

$$= 6.93 \times 10^{-3} M/s$$

$$(Q) r = \frac{-(4-2)}{40 \times 5} \times 10^{-2} M/s$$

$$r = \frac{r_A}{2} = 5 \times 10^{-3} M/s$$

$$(R) r = K[A]$$

$$= \frac{0.693}{80} \times 0.04$$

$$= 3.465 \times 10^{-4} M/s$$

$$(S) KA = \frac{1}{t_{1/2}} \times \frac{1}{[A]_0} = \frac{1}{40} \times \frac{1}{4/5}$$

$$r = \frac{5}{40 \times 4 \times 2} \times (0.08)^2 = 2.5 \times 10^{-5} M/s$$

PART-2 : CHEMISTRY

SECTION-II

4. Ans (1.00)

Elevation in boiling point Assuming ionic compound to be xy

$$\Delta T_b = i K_b m \text{ (molality)} \quad XY_{(s)} \rightarrow X^+ + Y^-$$

$$0.52 = i \cdot 0.52 \times \frac{\text{moles}}{1 \text{ kg water (solvent)}} \quad i = \text{Van'toff factor}$$

$$\text{moles of compound} = \frac{1}{2} \text{ mole} = 2$$

$$\frac{1}{2} \text{ mole compound} = 3 \times 10^{23} \text{ molecules as}$$

100% dissociation

1 unit cell like fcc (NaCl) contains 4 moles of ionic compound

$$\text{So, number of unit cells} = \frac{3 \times 10^{23}}{4}$$

$$r^- = \text{radius of anion} = 2 \text{ \AA}$$

$$r^+ = \text{radius of cation} = r^+ \text{ \AA}$$

$$\text{edge length} = 2(r^- + r^+) = 2(2 + r^+) \text{ \AA}$$

$$\text{Volume of solid cube} = (2.532)^3 \text{ cm}^3$$

$$\text{Volume of all the unit cell} = \underbrace{\frac{3}{4} \times 10^{23}}_{\text{No. of unit cell}} \times [2(2 + r^+)]^3$$

$$21.6 = 10^{23} \times (2(2 + r^+))^3$$

$$6 \times 10^{-8} \text{ cm} = 2(2 + r^+) \text{ \AA}$$

$$6 \times 10^{-8} \text{ cm} = 2(2 + r^+) 10^{-8} \text{ cm}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1.

Ans (A,B,D)

$$f(x) = \begin{cases} \pi \log_{\pi}(-x) & x < -\pi \\ -x & -\pi \leq x < 0 \\ x & 0 \leq x \leq \pi \\ \pi \log_{\pi} x & x > \pi \end{cases}$$

$\therefore f(x)$ is continuous everywhere

$\Rightarrow$ check differentiability at $x = \pi$

$$\Rightarrow \begin{cases} f'(\pi^+) = \pi \cdot \frac{1}{\pi \ln \pi} \\ f'(\pi^-) = 1 \end{cases} \quad \text{Not differentiable}$$

at $x = 0$

$$\begin{cases} f'(0^+) = 1 \\ f'(0^-) = -1 \end{cases} \quad \text{Not differentiable at } x = 0$$

at $x = -\pi$

$$\begin{cases} f'(-\pi^+) = -1 \\ f'(-\pi^-) = -\frac{1}{\ln \pi} \end{cases} \quad \text{Not differentiable at } x = 0$$

2.

Ans (D)

$$p'(x) = e^{(\ln x)^2} \cdot \left(\frac{2 \ln x}{x} \right)$$

$$\therefore p'(e^{-1}) = -2e^2$$

$$q'(x) = e^{x \ln(\ln x)} \cdot \left[\frac{1}{\ln x} + \ln(\ln x) \right]$$

$$q'(e) = 1$$

$$r'(x) = \frac{1 - \ln x}{x^2}, r'(e^2) = -\frac{1}{e^4}$$

$$\therefore \frac{p'(e^{-1}) \cdot r'(e^2)}{q'(e)} = \frac{2}{e^2}$$

3. Ans (A,B,C,D)

$$f''(x) = \lambda x$$

$$f'(x) = \frac{\lambda x^2}{2} + c \Rightarrow f'(1) = 0 \Rightarrow c = -\frac{\lambda}{2}$$

$$f'(x) = \frac{\lambda}{2}(x^2 - 1)$$

$$f(x) = \frac{\lambda}{2}\left(\frac{x^3}{3} - x\right) + c$$

$$f(0) = c = 1$$

$$f(x) = \frac{\lambda}{2}\left(\frac{x^3}{3} - x\right) + 1$$

$$f(1) = 2 \Rightarrow \lambda = -3$$

$$f(x) = \frac{-x^3}{2} + \frac{3x}{2} + 1$$

$$(A) f(x) - f(-x) = 6x - 10$$

$$\Rightarrow -x^3 + 3x = 6x - 10$$

$$x^3 + 3x - 10 = 0$$

(Number of solution is 1 as the function is

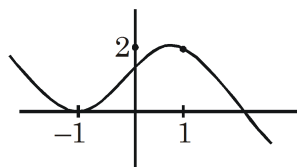
increasing)

$$(B) g(x) = \frac{x^3}{2} - \frac{3x}{2} - 1$$

$$g'(x) = \frac{3x^2}{2} - \frac{3}{2} \Rightarrow \frac{3}{2}(x-1)(x+1)$$

increasing of $g(x) : x \in (-\infty, -1] \cup [1, \infty)$

$$(C) f(x) = \frac{-x^3}{2} + \frac{3x}{2} + 1$$



$$f'(x) = -\frac{3}{2}(x-1)(x+1)$$

$$(D) p(\theta) = f(\sin\theta) \Rightarrow p(\theta) \in [0, 2]$$

4. Ans (A,B,C)

$$\sin x + \sin y = 1$$

$$\Rightarrow \cos x + \cos y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{\cos x}{\cos y}$$

$$\Rightarrow -\sin x - (\sin y) \left(\frac{dy}{dx}\right)^2 + (\cos y) \frac{d^2y}{dx^2} = 0$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{\sin x + \sin y \left(-\frac{\cos x}{\cos y}\right)^2}{\cos^2 y} = \frac{\cos^2 y \cdot \sin x + \sin y \cdot \cos^2 x}{(\cos y)^3}$$

$$\text{Put } \sin x = t, \sin y = 1 - t, \cos y = \sqrt{2t - t^2}$$

$$\lim_{x \rightarrow 0} x^\alpha \frac{d^2y}{dx^2} = \lim_{t \rightarrow 0} \frac{t^\alpha [(2t - t^2)t + (1-t)(1-t^2)]}{(\sqrt{2t - t^2})^3}$$

$$\text{exists} \Rightarrow \alpha \geq \frac{3}{2}$$

5. Ans (B,C)

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{\substack{x \rightarrow 1^+ \\ n \rightarrow \infty}} \frac{2 + \frac{x}{x^{2n}} + \frac{2}{x^{2n}}}{1 - \frac{x^2}{x^{2n}} + \frac{3}{x^{2n}}} = 2$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{\substack{x \rightarrow 1^- \\ n \rightarrow \infty}} \frac{2x^{2n} + x + 2}{x^{2n} - x^2 + 2} = \lim_{x \rightarrow 1^-} \frac{x + 2}{-x^2 + 2} = \frac{3}{1} = 3$$

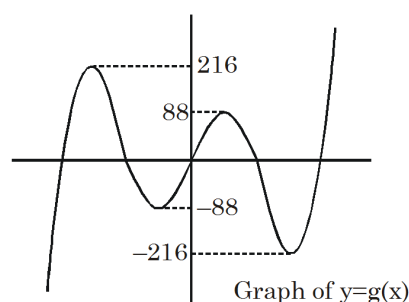
$\therefore \text{LHL} \neq \text{RHL}$

$\therefore$ Discontinuous at $x = 1$

Similarly

$$\lim_{x \rightarrow 2^+} f(x) = 2; \lim_{x \rightarrow 2^-} f(x) = 2$$

6. Ans (A,B,C)



$$\text{Let } g(x) = 3x^5 - 50x^3 + 135x$$

$$g'(x) = 0 \text{ for } x = \pm 1, \pm 3$$

$$g(1) = 88; g(-1) = -88; g(3) = -216; g(-3) = 216$$

Now consider $g(x) = a$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (A)**

$$(I) \phi(x) = 3f\left(\frac{x^2}{3}\right) + f(3-x^2)$$

$$\phi'(x) = 2x \left(f'\left(\frac{x^2}{3}\right) - f'(3-x^2) \right)$$

$$\text{Now } \frac{x^2}{3} - (3-x^2) = \frac{4x^2}{3} - 3 > 0$$

$$\Rightarrow x < -\frac{3}{2}, x > \frac{3}{2}$$

$$\frac{4x^2}{3} - 3 < 0 \Rightarrow -\frac{3}{2} < x < \frac{3}{2}$$

$$f'\left(\frac{x^2}{3}\right) - f'(3-x^2) \text{ is positive in } \left(-3, -\frac{3}{2}\right) \text{ and } \left(\frac{3}{2}, 4\right) \text{ and negative in } \left(-\frac{3}{2}, \frac{3}{2}\right) \text{ as } f'(x) \text{ is increasing } (\because f''(x) > 0)$$

$$\therefore \phi'(x) \text{ is negative in } \left(-3, -\frac{3}{2}\right), \left(0, \frac{3}{2}\right)$$

$$\text{i.e. decreasing in } \left[-3, -\frac{3}{2}\right], \left[0, \frac{3}{2}\right] \text{ and}$$

$$\text{increasing in } \left[\frac{3}{2}, 4\right], \left[-\frac{3}{2}, 0\right]$$

$$(II) g(x) = (a^2 - 3a + 2)\cos\frac{x}{2} + (a-1)x + \sin 1$$

$$g'(x) = -\frac{1}{2}(a^2 - 3a + 2)\sin\frac{x}{2} + (a-1)$$

$$= (a-1)\left(1 - \frac{1}{2}(a-2)\sin\frac{x}{2}\right)$$

$$g'(x) = 0 \text{ when } a = 1 \text{ or } \sin\frac{x}{2} = \frac{2}{a-2}, a \neq 2$$

$$\therefore \left|\frac{2}{a-2}\right| < 1 \Rightarrow a \in (-\infty, 0) \cup (4, \infty)$$

$\Rightarrow$ It possess critical points

if $a \in (-\infty, 0) \cup [4, \infty) \cup \{1\}$

8. **Ans (C)**

(P) Given equation holds good for $x = -1$ only

(Q)

$$a = \lim_{x \rightarrow \infty} 5 \left(2 \left(\frac{3}{5} \right)^x + 3 \left(\frac{4}{5} \right)^x + 4 \right)^{1/x} = 5(0+0+4)^0 = 5$$

$$b = \lim_{x \rightarrow (-\infty)} 3 \left(2 + 3 \left(\frac{4}{3} \right)^x + 4 \left(\frac{5}{3} \right)^x \right)^{\frac{1}{x}} = 3$$

(R) At $x = n, n \in I$

$$\text{LHL} = n \sin(\pi(2n - (n-1)))$$

$$= n \sin((n+1)\pi) = 0$$

$$\text{RHL} = f(n) = n \sin n\pi = 0$$

$\therefore f(x)$ is always continuous

$$(S) \lim_{x \rightarrow 0} 3 \left(\frac{e^{x/3} - \frac{x}{3} - 1}{\frac{x^2}{9} \cdot 9} \right) = \frac{3}{9} \cdot \frac{1}{2} = \frac{1}{6}$$

9. **Ans (A)**

$$(P) g'(0) = f'(0) \cdot (-1 + 0 + f(0)) = 0$$

$$(Q) \text{ take log } \Rightarrow y(1 + \ln x) = x$$

$$\Rightarrow y = \frac{x}{1 + \ln x}$$

Now differentiate

$$(R) -1 \leq x \leq 2 \Rightarrow -2 \leq 2x \leq 4$$

$\therefore$ point of discontinuity

when $2x = -1, 0, 1, 2, 3, 4$

$$\text{i.e. } x = -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, 2$$

$$(S) I = e^{\lim_{x \rightarrow 0} (1 + \tan x - 1) \cdot \frac{1}{\sin x}} = e^1$$

10. **Ans (B)**

$$\sin^{-1}x + \cos^{-1}ay + \cos^{-1}(bxy) = \frac{\pi}{2}$$

$$\cos^{-1}(ay) + \cos^{-1}(bxy) = \cos^{-1}x$$

$$\cos(\cos^{-1}(ay) + \cos^{-1}(bxy)) = \cos(\cos^{-1}x)$$

$$(ay)(bxy) - \sqrt{1-a^2y^2} \cdot \sqrt{1-b^2x^2y^2} = x \quad \dots(1)$$

(P) $a = 1, b = 2$

$$\Rightarrow (y)(2xy) - \sqrt{1-y^2} \cdot \sqrt{1-4x^2y^2} = x$$

$$\Rightarrow (2xy^2 - x)^2 = (1-y^2)(1-4x^2y^2)$$

$$\Rightarrow (1-x^2)(1-y^2) = x^2y^2$$

(Q) $a = 2, b = 1$

$$\Rightarrow (2xy^2 - x)^2 = (1-4y^2)(1-x^2y^2)$$

$$\Rightarrow x^2 + 4y^2 = 1 + 3x^2y^2$$

$$(R) a = 0, b = 1 \Rightarrow -\sqrt{1 - x^2 y^2} = x$$

$$\Rightarrow x^2 = 1 - x^2 y^2$$

$$\Rightarrow 2x = -x^2 2yy' - 2xy^2$$

$$\Rightarrow y' = -\frac{y^2 + 1}{xy}$$

$$(S) \text{ if } a = b \quad x^2 + a^2 y^2 - x^2 y^2 a^2 - 1 = 0$$

$$\Rightarrow x = \pm 1 \text{ or } y = \pm \frac{1}{a}$$

$$\Rightarrow \text{area of rectangle} = 2 \cdot \frac{2}{a} = 1$$

$$\Rightarrow a = 4 = b$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (3.00)**

$$f(x) = \begin{cases} ax & ; x < 2 \\ ax^2 - bx + 3 & ; x \geq 2 \end{cases}$$

for $f(x)$ to be continuous $\forall x \in \mathbb{R}$ it should be continuous at $x = 2$

$$f(2^-) = \lim_{x \rightarrow 2^-} ax = 2a$$

$$f(2^+) = f(2) = 4a - 2b + 3$$

$$f(2^-) = f(2) = f(2^+) \Rightarrow 2a = 4a - 2b + 3$$

$$2a - 2b + 3 = 0 \quad \dots(1)$$

$$f'(x) = \begin{cases} a & ; x < 2 \\ 2ax - b & ; x > 2 \end{cases}$$

$f(x)$ will be differentiable $\forall x \in \mathbb{R}$ if it is differentiable at $x = 2 \Rightarrow f'(2^-) = f'(2^+)$

$$\Rightarrow a = 4a - b$$

$$\Rightarrow 3a - b = 0 \quad \dots(2)$$

$$\text{By (1) \& (2) } a = \frac{3}{4} \text{ and } b = \frac{9}{4}$$

$$\therefore a + b = 3$$

2. **Ans (3.20)**

$$f(x) = (x-1)^{2019} + (x-2)^{2019} + (x-3)^{2019} + \dots + (x-2019)^{2019}$$

$f(x)$ is an increasing function so it has only one real root

$$x = \frac{1 + 2019}{2} = \frac{2020}{2} = 1010$$

$$\frac{\lambda}{315} = \frac{1010}{315} = 3.20$$

3. **Ans (2.00)**

$$\tan(\cos^{-1}(\sin(\cos^{-1}x)))$$

$$= \tan\left(\frac{\pi}{2} - \sin^{-1}\left(\sin\left(\frac{\pi}{2} - \sin^{-1}x\right)\right)\right)$$

$$= \cot(\sin^{-1}(\cos(\sin^{-1}x)))$$

$$\Rightarrow f(x) = 1$$

$$\sum_{r=0}^{\infty} \frac{1}{2^r} = \frac{1}{1 - \frac{1}{2}} = 2$$

4. **Ans (0.50)**

$$\lim_{n \rightarrow \infty} \prod_{r=1}^{r=n-1} \left(1 - \frac{1}{(r+1)^2}\right)$$

$$\prod_{r=1}^{r=n-1} \frac{(r+1)^2 - 1}{(r+1)^2} = \prod_{r=1}^{r=n-1} \frac{r}{(r+1)} \frac{(r+2)}{r+1}$$

$$= \prod_{r=1}^{r=n-1} \frac{r}{r+1} \prod_{r=1}^{r=n-1} \frac{r+2}{r+1}$$

$$= \frac{1}{n} \frac{(n+1)}{2}$$

$$\lim_{n \rightarrow \infty} \prod_{r=1}^{r=n-1} \left(1 - \frac{1}{(r+1)^2}\right)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \frac{(n+1)}{2} = \frac{1}{2}$$

5. **Ans (9.00)**

$$f'(0^-) = \lim_{h \rightarrow 0} \frac{\left(\frac{(0-h)(\ell_n(1-h) + \ell_n(1+h))}{(\sec h - \cosh)} - 0\right)}{-h}$$

$$f'(0^-) = \lim_{h \rightarrow 0} \frac{\ell_n(1-h^2)}{(\sec h - \cosh)}$$

$$= \lim_{h \rightarrow 0} \frac{\ell_n(1-h^2)}{-h^2} \frac{\cosh}{\frac{1-\cosh}{h^2}} \times \frac{1}{(1+\cosh)} \times \frac{-h^2}{h^2}$$

$$= -1$$

$$f'(0^+) = (p^2 - 3p - 1)\cos 0 + 2 \times 0$$

$$= (p^2 - 3p - 1)$$

$$f'(0^-) = f'(0^+)$$

$$p^2 - 3p - 1 = -1$$

$$p = 0, 3 \Rightarrow p_1 = 0 \text{ \& } p_2 = 3$$

6. **Ans (0.20)**

Let

$$P(x) = a(x-1)^4 + b(x-1)^3 + c(x-1)^2 + d(x-1) + e$$

$$P(1) = e = 1$$

$$P'(1) = d = 0$$

$$P''(1) = 2c = 2 \Rightarrow c = 1$$

$$P'''(1) = 3.2.b = -12 \Rightarrow b = -2$$

$$P''''(1) = 4.3.2.1.a = 24 \Rightarrow a = 1$$

$$\therefore P(x) = (x-1)^4 - 2(x-1)^3 + (x-1)^2 + 1$$

$$\therefore P(2) = 1 \text{ \& } P(3) = 5$$

$$\therefore \frac{P(2)}{P(3)} = \frac{1}{5} = 0.2$$

 7. **Ans (8.00)**

 Eliminating t will give

$$y^2(x-1) = 1$$

 equation of tangent at $(2, 1)$ is $x + 2y = 4$

solving with curve,

$$\text{we get } x = 5 \text{ and } y = -\frac{1}{2}$$

$$Q = \left(5, -\frac{1}{2}\right) \text{ or } |PQ| = \frac{3\sqrt{5}}{2}$$

 8. **Ans (3.75)**

$$f\left(\frac{2x-3}{x-2}\right) = 5x-2$$

$$\frac{2x-3}{x-2} = f^{-1}(5x-2)$$

$$x = 3 \quad f^{-1}(13) = 3$$